Sarthak Dubey

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EDUCATION

Northeastern University, Boston, MA **Sept 2025 - Dec 2027**
Master's in Robotics GPA: 3.67/4.0
Relevant Coursework: Robotics Mechanics, Robot Dynamics, Control Systems, Mobile Robots, Reinforcement Learning, Natural Language Processing

Vellore Institute of Technology, Bhopal, India **Sept 2020 - April 2024**
Bachelor of Technology in Electronics & Communication Engineering (AI & Cybernetics) | CGPA: 7.9/10
Relevant Coursework: Embedded Systems, Microcontrollers, Digital Electronics, Signal Processing, Control Theory, Data Structures & Algorithms

SKILLS

- **Programming**: Python, C++, MATLAB, Linux/Ubuntu
- **Robotics & Middleware**: ROS1 (Noetic), ROS2 (Humble, Jazzy), RViz, Sensor Integration
- **Libraries & Frameworks**: OpenCV, PCL, CasADi, Pinocchio
- **Simulation**: Gazebo (Classic, Harmonic), MuJoCo, IsaacSim, IsaacLab
- **Hardware & Embedded Systems**: Arduino, Raspberry Pi, IMU, GPS, Motion Control Systems
- **Design & Prototyping**: SolidWorks, Rapid Prototyping, 3D Printing
- **Certifications & Training**: AWS(Cloud Foundations) Certification, VLSI Design Methodologies.

WORK EXPERIENCE

Kitolit Technologies, Hyderabad, India **Jun 2024 – Oct 2024**
Robotics Intern

- Developed indoor mobile robot prototypes using Raspberry Pi, Arduino, IMU, encoders, ultrasonic sensors, and DC motors to support autonomous navigation and interactive demonstrations.
- Implemented UART/I2C sensor integration and an OpenCV vision pipeline for real-time perception and obstacle avoidance, improving sensing reliability across changing indoor conditions.
- Designed and tested robot chassis, sensor mounts, and FDM components in SolidWorks while logging camera, IMU, and encoder data to diagnose hardware and navigation failures.

PROJECTS

SNGR: Selective Non-Gaussian Refinement for Ambiguous SLAM Factor Graphs
Link: https://arxiv.org/abs/2604.22065 | Python, GTSAM, iSAM2, Factor Graphs, Nested Sampling

- Developed a selective non-Gaussian SLAM framework by augmenting iSAM2 with targeted nested sampling to refine ambiguous regions where Gaussian posterior assumptions fail.
- Designed a failure-detection trigger using condition numbers of joint marginal covariance matrices to identify unreliable local regions before applying computationally expensive refinement.
- Implemented local refinement using the full nonlinear factor-graph likelihood and a gating mechanism to improve ambiguous estimates while preventing degradation in multimodal cases.
- Evaluated SNGR on range-only SLAM with incorrect data associations, achieving high-precision failure detection and consistent local likelihood improvements at lower cost than exhaustive non-Gaussian inference.

Multi-Sensor Fusion for Robot Localization
Project Report: Link | ROS2, C++, Python, EKF, Visual Odometry, GPS/IMU Fusion

- Designed a ROS2-based multi-sensor localization pipeline integrating u-blox GPS, IMU, and monocular camera streams through timestamp synchronization and TF frame transforms, enabling consistent state estimation across indoor–outdoor robot trajectories.
- Implemented IMU-aided visual odometry using quaternion-based orientation correction and camera pose fusion, reducing cumulative trajectory drift from 45.8% to 3.1% during extended navigation runs.
- Developed and benchmarked Complementary Filter and Extended Kalman Filter estimators in Python and ROS2 using covariance and innovation-residual analysis, evaluating localization robustness under sparse and intermittent GPS measurements.
- Implemented SVD-based Procrustes alignment and trajectory scale correction for visual odometry outputs, maintaining frame consistency and recovering localization accuracy after sensor dropouts and estimator reinitialization.

SignSync: Seamless ASL to English
Project Report: Link | Python, OpenCV, MediaPipe, LSTM, TensorFlow/Keras

- Developed a real-time computer vision pipeline using OpenCV and MediaPipe for hand and body pose estimation, extracting spatial keypoints and temporal motion features from live camera streams for dynamic gesture recognition.
- Built a Python-based data preprocessing pipeline for landmark normalization, sequence segmentation, and labeled dataset generation, converting tracked keypoints into structured NumPy tensors for supervised temporal learning.
- Trained and tuned an LSTM sequence-classification model in TensorFlow/Keras using dropout regularization and class balancing, improving recognition consistency across users and temporally similar ASL gestures.
- Integrated the perception and inference pipeline with a Django REST backend and Bluetooth-enabled Arduino OLED interface, enabling low-latency translation of recognized gestures into readable embedded-system output.

ABU Robocon 2022 - Competition Robot Design & Integration

- Designed robot chassis, structural components, and mechanical mounting assemblies in SolidWorks, using ANSYS-based stress analysis to evaluate load distribution and structural reliability during repeated competition testing and high-impact operation.
- Integrated LiDAR, encoders, motors, controllers, Arduino, and Raspberry Pi hardware with ROS-based control software, combining perception, actuation, and embedded systems into a functional competition robot platform.
- Fabricated, assembled, and iterated mechanical and electrical subsystems through machining, calibration, and full-system testing, resolving alignment, tolerance, and hardware-integration issues to improve robot reliability and repeatability.